International Journal of Heat and Mass Transfer 54 (2011) 5574–5587

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ijhmt

The thermal response of an infinite line of open loop wells for ground coupled heat pump systems

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ARTICLE INFO

Article history:
Received 25 June 2010
Accepted 6 July 2011
Available online 22 August 2011

Keywords:
Thermal storage
Ground coupled heat pump
Geothermal well
Standing column well
Well field
Analytical model

ABSTRACT

Ground thermal energy storage is a means of storing thermal energy underground during the summer and utilizing it during the winter. The main use of such a technology is in the heating, ventilating and air conditioning sector where the ground provides a stable temperature reservoir for a heat pump system. Heat pumps are mechanical systems that provide heating to a space in the winter, and cooling in the summer. They are increasingly popular because the same system provides both heating modes, depending on the direction of the cycle upon which they operate. The stable temperature reservoir allows the heat pump system to run at a higher efficiency. Thermal energy is transmitted to and from the ground by circulation of water through standing column geothermal wells. In commercial applications, the ground storage is enabled by a field of multiple wells; hence proper well spacing is important for optimum performance for a fixed land area and required storage capacity. If wells are too closely spaced, one system may thermally encroach on another thereby lowering the efficiency of both.

This paper presents a comprehensive analysis of the thermal response of a line of evenly spaced wells, operating concurrently. A numerical model is developed, validated using empirical data, and compared to an approximate analytical model that allows for the assessment of coefficient of performance degradation. The analytical model is solved by separating the problem into three time regimes. First, for small values of time, heat is absorbed predominantly by the thermal capacity of the well water. Second, for intermediate times, heat flow through an infinite cylinder dictates the thermal response. Third, for late times, planar conduction dominates due to thermal encroachment between adjacent wells. The transition times between these regimes are derived from characteristic times distinct to the driving physics. Using the analytical model, the temperature response of the well in a line is compared to an isolated well showing the degradation in well performance with time. This degradation in performance is then related to the coefficient of performance of a reversible heat pump system to assess the impact on efficiency. The results quantify the decrease in coefficient of performance of a ground coupled heat pump system sourced by a line of wells as a function of time and decreasing well spacing. The coefficient of performance figure is a figure of merit for geothermal well field designers to properly assess the financial benefits of a ground coupled heat pump system compared to conventional systems.

1. Introduction

Heat pumps are mechanical systems that operate in a thermodynamic cycle to pump heat from low temperature to high temperature. The advantage of using ground source heat pumps is that the ground water temperatures throughout the year are more stable than the air temperatures used in conventional heat pumps. This leads to increases in efficiency. A second major advantage of ground source systems is that energy extracted from structures during the cooling season can be stored in the ground and extracted during the heating season.

Geothermal wells are of great interest in the building heating and cooling industry as the preferred method for utilizing the ground as a thermal reservoir instead of the atmosphere. There are many types of geothermal wells used with ground source heat pumps including hybrid systems that integrate solar energy inputs [1–3]. The geothermal wells studied in this paper are referred to as standing column wells (SCW). A typical standing column well is between 100–500 m deep and 0.1–0.25 m (4–10 in.) in diameter.

Standing column wells operate in an open loop and the circulating water is physically in direct contact with the ground and groundwater. Water is extracted from below the water table into a building, circulated through the building heating, ventilation and air conditioning (HVAC) system, and returned to the ground. A schematic of such a system is shown in Fig. 1. The ground surrounding the well is specific to the region and ranges from porous

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doi:10.1016/j.ijheatmasstransfer.2011.07.027

Nomenclature

A_c	cross-sectional area (m^2)
Co	Courant number
COP	coefficient of performance
C_p	heat capacity (J/kg K)
D_{ip}	diameter (m)
D_h	hydraulic diameter (m), 4·(cross-sectional area)/perimeter
Fo_i	Fourier number associated with the numerical model (i = 1, 2, 3, 4 defined in paper)
Fo_L	Fourier number with the length between wells as the characteristic length
Fo_R	Fourier number with the well radius as the characteristic length
H	depth of the well (m)
h	convective heat transfer coefficient (W/m^2 K)
k	thermal conductivity (W/m K)
L	distance between wells (m)
L_f	distance between wells plus the added length associated with a square well (m)
l	addition length associated with a square well (m)
m	flow rate through the well (kg/s)
N_w	number of wells to which the building heat load is applied
NTU_{1d}	number of transfer units in annular region to the ground formation
NTU_{1u}	number of transfer units from the return pipe to the annular region
NTU_{2d}	number of transfer units from the annular region to the return pipe
NTU_g	number of transfer units from the annular region to the ground formation
NTU	number of transfer units from the annular region to the return pipe
Nu	Nusselt number
Pr	Prandtl number
Q	heat load applied to the well (W)
q_r	heat flux for radial heat flow (W/m^2)
q_y	heat flux for planar heat flow (W/m^2)
Re	Reynolds number
R_o	radius of the well (m)
R_{pipe}	total resistance from the annulus flow to the return pipe flow (mK/W)
T_g	temperature of the ground formation (°C)
T_i	initial ground formation temperature (°C)
T_r	temperature of the return flow through the polypipe (°C)
T_w	temperature of the flow down the annulus of the well (°C)
t_{HT}	time associated with heat up of the well using Courant number (s)
t_R	for thermal front to propagate to the plane of symmetry between wells (s)
$V_{annulus}$	volume of water inside the annulus (m^3)
V_{pipe}	volume of the water inside the return pipe (m^3)

Greek symbols

α_g	thermal diffusivity of the ground (m^2/s)
γ	Euler's constant ≈ 0.577
ΔT_G	temperature response due to thermal resistance in radial system (°C)
ΔT_P	temperature response due to thermal resistance in planar system (°C)
Δx_i	node spacing at node (i, j, k) in the x-direction (m)
Δy_j	node spacing at node (i, j, k) in the y-direction (m)
Δz_k	node spacing at node (i, j, k) in the z-direction (m)
ρ_g	effective density of the ground formation (kg/m^3)
ρ_w	density of water (kg/m^3)
θ	non-dimensional temperature, Eq. (42)
μ_w	viscosity of water (kg/m s)

Subscripts and superscripts

d	down the annulus
d, k	down the annulus at node k
g	ground formation
H	heating mode
i	node number in the x-direction
ip	return pipe inner wall
j	node number in the y-direction
k	node number in the z-direction
o	outer wall of well
op	return pipe outer wall
p	pipe
R	refrigeration mode
u	up the return pipe
u, k	up the annulus at node k
w	water

rock to solid rock with random cracks, allowing for advection through the ground formation. The capacity of the ground for storing energy is determined by its thermal diffusivity, consisting of its thermal conductivity, density, and specific heat. Additionally, groundwater flow advection affects the spatial distribution of temperature and therefore affects energy storage. The ground formation properties are generally anisotropic, and the temperature profile is largely two-dimensional and transient.

Standing column wells have traditionally been designed following water well technology that has been extensively studied in the past century and encoded in well accepted design handbooks [4,5]. Circulating SCWs exhibit physics similar to the physics encountered in petroleum and geothermal wells during drilling. The major difference between these wells and standing column wells is that the goal of drilling is to reach an oil reservoir or plume of hot rock, and the loss of petroleum or heat to the ground formation during extraction is undesirable. In contrast, standing column well operation is dependent on good thermal interaction with the ground formation. The design of petroleum wells is reasonably standardized [6]. The length and diameter of oil wells are orders of miles and feet, respectively. In groundwater technology, the length and diameter of wells is limited to hundreds of feet, and inches, respectively. Groundwater well technology has been extensively studied in the ground water hydraulics literature [4] and ground-coupled (closed loop) geothermal systems in the HVAC literature [7,8].

Standing column wells have attracted more attention in recent years due to efficiency gains over closed loop systems: Deng [1], Rees et al. [9], and Spitler et al. [10] focused on single SCW modeling using both analytical and numerical techniques that assumed isotropic "effective" properties. These studies found that the defining properties that limit operation are the ground thermal conductivity, bleed rate, and well depth. Bleed rate is the intentional removal of water from a well to induce groundwater movement and enhance thermal transport. The advection through the ground formation due to pressure gradients is especially important in determining the thermal interaction with the well.

There has been little fundamental work reported beyond the study of single isolated circulating standing column wells. Most SCW installations for large systems require the utilization of a field of wells spaced at some distance from each other. A typical well field may consist of a matrix of evenly spaced wells extending in two directions. For practical reasons it is desirable to place the wells as close to each other as possible, so that the required yearly storage capacity can be accomplished in a minimal land area. However, wells must be properly spaced to avoid thermal interference and thereby reduction in their ability to store energy. As a first step towards modeling the problem, this paper presents the development of an approximate analytical model, and a validated numerical model for determining the thermal behavior of a line of evenly spaced wells operating concurrently. The numerical model is validated using empirical data while the approximate analytical model is validated against the numerical model. The approximate analytical model provides a means for calculating the degradation of efficiency due to thermal encroachment and its effect on the ideal coefficient of performance for a ground-coupled heat pump system.

2. Analysis of problem

Fig. 1 illustrates the typical operation of a standing column well in an infinite line of wells. Water from the building cooling/heating system circulates down the annulus and returns up the return pipes inserted into the well. Heat is exchanged between the annulus and the formation and between the annulus and the return flow. The total heat load per well, Q, is obtained by adding the total work input, W, with the total heat absorbed by the building HVAC system, Q_{Building}, and dividing by the number of wells, N_w; assuming each well dissipates an equal amount of the total load. The governing thermal energy equation for flow down the annulus is a one-dimensional transient formulation for the local mixed mean fluid temperature that incorporates heat transfer from the annulus flow to the ground formation and from the annulus flow to the return flow. Constant properties are assumed yielding:

$$\frac{\rho_w A_{cd}}{m}\frac{\partial T_w(z,t)}{\partial t} = -\frac{2\pi h_o R_o}{mc_{pw}}(T_w(z,t) - T_g(R_o,z,t) - 2 \times \frac{(T_w(z,t) - T_r(z,t)}{mC_{pw}R_{\text{pipe}}} - \frac{\partial T_w(z,t)}{\partial z} \quad (1)$$

h_o is the convective coefficient from the water in the annulus to the wall of the well. Similarly, the energy equation for flow up the return pipe is a one-dimensional transient formulation for the mixed mean temperature in the axial flow with heat transfer between the pipe flow and the annulus flow,

$$\frac{\rho_w A_{cu}}{m}\frac{\partial T_r(z,t)}{\partial t} = \frac{2(T_w(z,t) - T_r(z,t))}{mC_{pw}R_{\text{pipe}}} - \frac{\partial T_r(z,t)}{\partial z} \quad (2)$$

The factor "2" in Eqs. (1)–(3) is specific to the present analysis because there were two return pipes in the well analyzed. This number should be neglected for different orientations. R_{pipe} is the total thermal resistance from the annulus fluid to the return pipe fluid and is used to find the number of transfer units, NTU; a typical heat exchanger parameter:

$$NTU = \frac{2H(R_{\text{pipe}})^{-1}}{mC_{pw}} = \frac{2H\left[\frac{1}{\pi h_{ip}D_{ip}} + \frac{\ln(D_{op}/D_{ip})}{2\pi k_p} + \frac{1}{\pi h_{op}D_{op}}\right]^{-1}}{mC_{pw}} \quad (3)$$

The annulus and return pipes are not only thermally connected but also physically connected in the axial direction by two boundary conditions for the top ($z = H$) and bottom ($z = 0$) of the well given respectively by,

$$T_w(H,t) - T_r(H,t) = \Delta T_{\text{HP}} = \frac{Q}{mC_{pw}} \quad (4)$$

and

$$T_w(0,t) = T_r(0,t) \quad (5)$$

where Q is the heat load per well, m is the mass flow rate per well, and ΔT_{HP} is the change in temperature across the heat pump.

For early times, the thermal load is mostly absorbed by displacing the initial thermal mass of the water in the well with the thermal mass of the water leaving the building. When the well becomes thermally saturated, the thermal load is then predominantly absorbed by the ground formation. The approximate time for this initial thermal saturation is the residence time of the well,

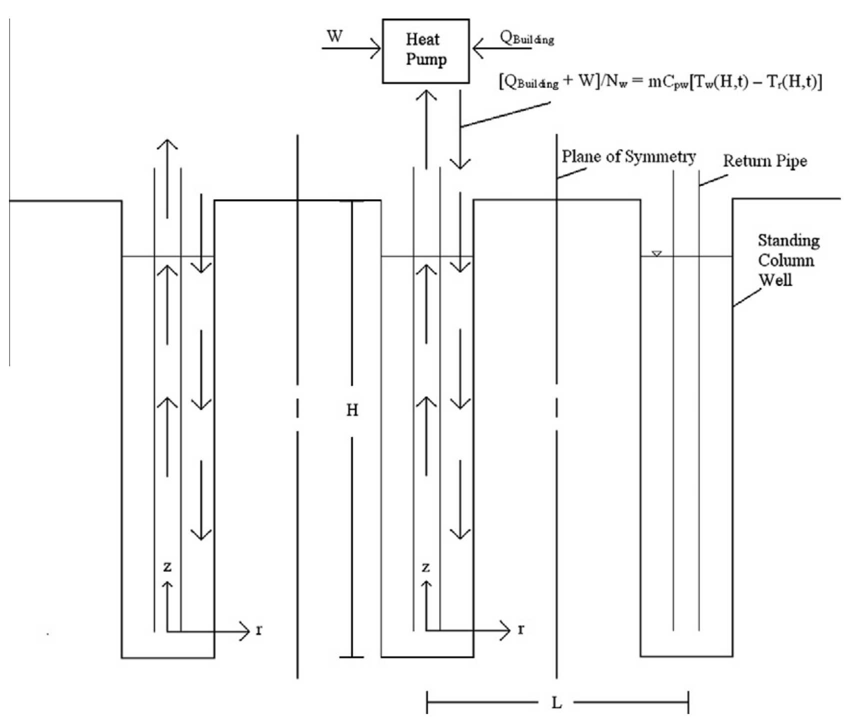

Fig. 1. Diagram of standing column well operation in an infinite line of wells.

defined as the time it takes to completely displace the initial mass of water in the well:

$$t_{\mathrm{HT}} = \left(\frac{\rho_w V_{\mathrm{pipe}} + \rho_w V_{\mathrm{annulus}}}{m}\right) \tag{6}$$

The ground formation is modeled as a one-dimensional radial semi-infinite solid, where conduction in the axial direction is neglected due to scaling,

$$\frac{\partial T_g(r,z,t)}{\partial t} = \alpha_g \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T_g(r,z,t)}{\partial r}\right) \tag{7}$$

Here α_g is the effective thermal diffusivity given by $k_g/(\rho_g \cdot C_{pg})$. The ground formation is modeled using "effective" properties derived from empirical data. This method is consistent with the work of the Spitler group [1,9,10]. In this approach, the thermal conductivity, k_g, is not the thermal conductivity of the ground formation but rather the "effective" thermal conductivity which is an approximation of the effects of conduction and advection through both the rock and the water in the rock pores and cracks. The boundary equation for the annulus to the ground formation is

$$-k_g \frac{dT_g(R_o,z,t)}{dr} = h_o(T_w(z,t) - T_g(R_o,z,t)) \tag{8}$$

The farfield boundary condition for the semi-infinite solid is

$$T_g(\infty,z,t) = T_i \tag{9}$$

Here T_i is assumed to be constant in the axial direction. Deng et al. [11] reports that for ground thermal gradients less than 0.6 °C/100 m, the vertical thermal gradient may be neglected. This assumption is adopted in the present paper.

2.1. Cartesian domain and limits of analysis

When the thermal fronts of two adjacent wells interact, the problem loses its radial symmetry. As the spacing between wells becomes small, the problem transitions to a planar half space problem where the line of wells imposes a constant heat flux boundary condition onto a half space. Fig. 2 depicts the two limiting cases. The "isolated well limit" results in purely 1D radial conduction whereas the "zero spacing limit" results in purely planar 1D conduction at some distance from the well. The transition time from radial to planar conduction is approximated using the well known analytical solution for the thermal profile of a one-dimensional semi-infinite radial solid with constant heat flux on an inner radius, R_o, from Carslaw and Jaegar [12],

$$T_g(r,t) - T_i = \frac{Q}{4\pi H k_g}\ln\left(\frac{4\alpha_g t}{\exp(\gamma) r^2}\right) \tag{10}$$

The transition time (t_R) is estimated as the time for the radial temperature front to reach the symmetry line between wells; therefore assuming $r = L/2 - R_o$ and setting the left side of Eq. (10) to zero gives the time when the thermal front interacts with the plane of symmetry.

$$t_R = \frac{\exp(\gamma)\left(\frac{L}{2} - R_o\right)^2}{4\alpha_g} \tag{11}$$

After this approximate transition time, the governing equation for thermal flow can be modeled by a two-dimensional transient Cartesian equation,

$$\frac{\partial T_g(x,y,z,t)}{\partial t} = \alpha_g\left(\frac{\partial^2 T_g(x,y,z,t)}{\partial x^2} + \frac{\partial^2 T_g(x,y,z,t)}{\partial y^2}\right) \tag{12}$$

Again α_g is the effective thermal diffusivity based on effective formation properties. Fig. 3 shows the transition from radial coordinates to planar coordinates as time progresses. It is important to note that at large times, the near well thermal problem remains radial while in the farfield it becomes planar. It will be shown that adopting a Cartesian system simplifies the numerical model, even though it is not optimal for the well near field. The cross-section of the domain to be modeled is shown in Fig. 4. Here, the well is modeled with an equivalent square cross-section to simplify the numerical grid. Because of symmetries, it is necessary to model only a quarter section. The boundary conditions for this domain, also shown in Fig. 4, are a symmetry boundary condition at the plane of symmetry between adjacent wells where $x = L/2$, a symmetry boundary condition at $x = 0$, a symmetry boundary condition at $y = 0$, and a farfield boundary condition at $y \to \infty$.

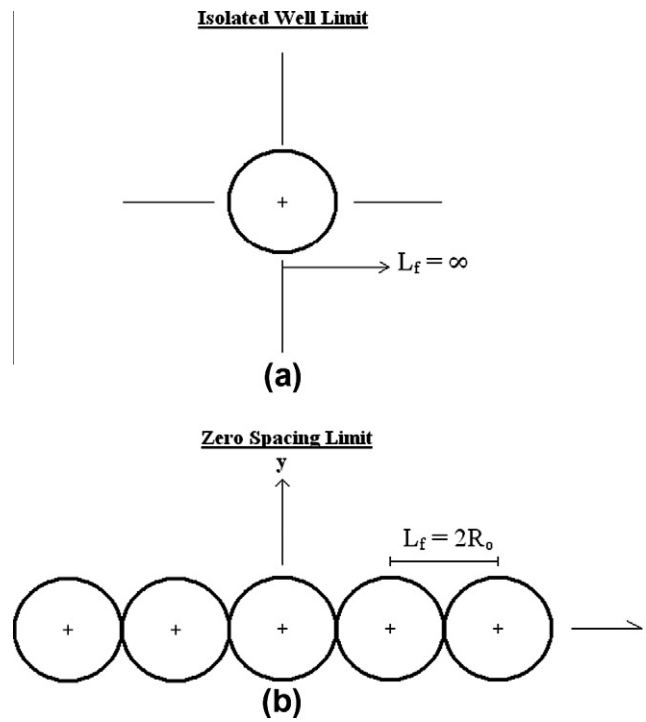

Fig. 2. (a) Isolated well limit where the length between wells, $L_f = \infty$. (b) Zero spacing limit where the length between wells, $L_f = 2R_o$.

3. Analysis of heat transfer constants

The heat transfer coefficients h_o, h_{ip}, and h_{op} in Eqs. (1)–(3), are determined from appropriate correlations for laminar and turbulent flow in a circular pipe, h_{ip}, and in a circular pipe annulus, h_{op}, and h_o. Empirical data provide the "effective" properties of the ground. Calculation of the convective heat transfer coefficients depends on the conditions of the problem. The Reynolds number is defined as,

$$Re = \frac{\rho_w V D_h}{\mu_w} \tag{13}$$

The Nusselt number with respect to axial length is calculated using Tables 8–11 and 8–13 from Kays and Crawford [13] for constant heat rate, which incorporate the thermal and hydrodynamic entry region heat transfer coefficient. For return pipe laminar flow, Table 8–13 from Kays and Crawford [13] is used to estimate the heat transfer coefficient. Most SCWs operate in the turbulent flow regime $2100 < Re < 100{,}000$. According to [13] Table 14-3 and Fig. 14-7, the thermal entry region has little influence on Nusselt number for fluids with Prandtl numbers between 0.7 and 10 and Reynolds numbers of 100,000. Hence, an approximate convective

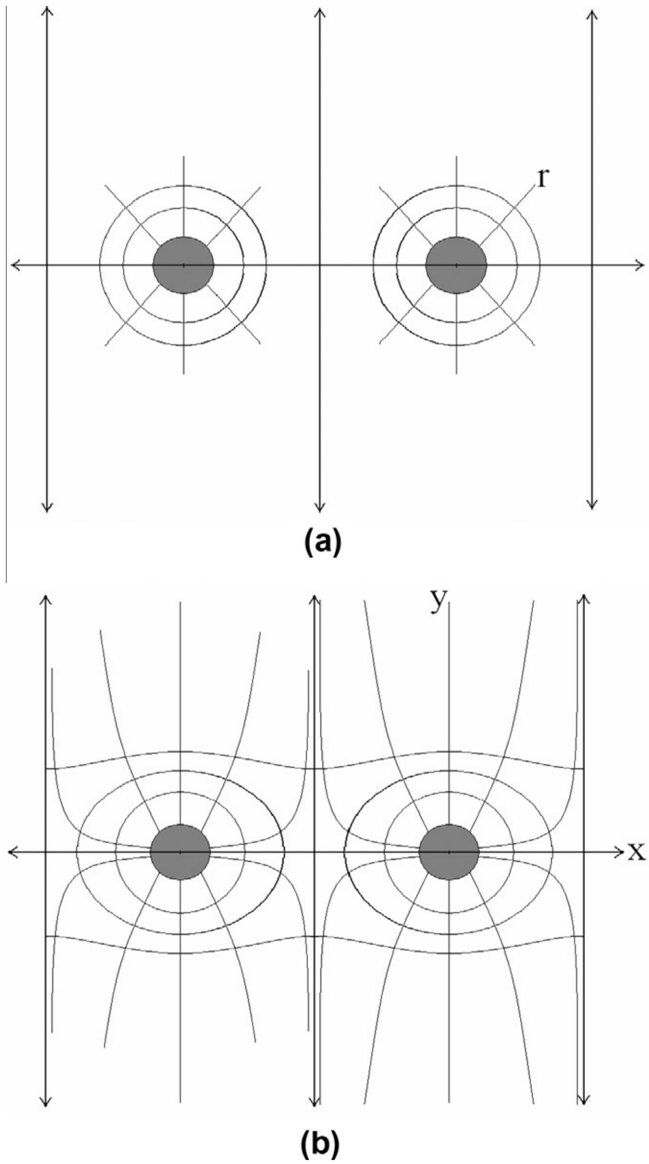

Fig. 3. (a) Temperature isotherm contours for $t_{HT} < t < t_R$ where the problem is radially dominated. (b) Temperature isotherm contours for $t \geqslant t_R$ where the problem has transitioned to be a planar problem in the farfield.

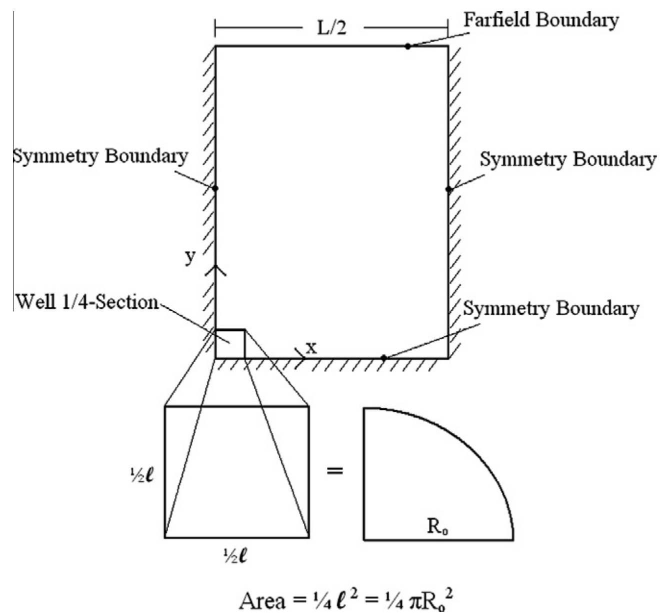

Fig. 4. Boundary conditions of planar domain including the relation of the square well to a circular well.

heat transfer coefficient is obtained using [13] Tables 14-3 and 14-4 for turbulent flows.

It is noteworthy that for typical operating conditions of SCWs, the thermal resistances due to the convective heat transfer coefficients are negligible compared to the conduction resistances provided by the return pipe and the ground formation, hence extreme accuracy in the specification of the heat transfer coefficient is not warranted. For the conduction resistances, the return pipe thermal conductivity, k_p, is obtained from the manufacturer, and the ground formation "effective" thermal conductivity, k_g, is obtained through empirical data reduction as demonstrated in the following section.

3.1. Experimental determination of formation effective thermal conductivity

An experiment was performed on a geothermal well used to provide heating and cooling to a residence hall at the authors' institution [14]. The well is 800 ft (243 m) deep and 6 in. (0.152 m) in diameter. The flow rate was 32 GPM (2.0186 kg/s), heating load was 213,000 BTUH (63 kW), and pump power input was 1.5 hp (1.12 kW). The water table is 70 ft (21 m) below ground level and the ground composition is predominantly Gneiss.

The effective conductivity was calculated using a linear regression estimation method, a standard method for calculating thermal conductivity [15]. From Eq. (10), the derivative with respect to the natural logarithm of time is a constant at large times, given by

$$\frac{dT}{d\ln(t)} = \frac{Q}{4\pi H k_g} \tag{14}$$

Using a linear best fit to find the slope of the semi-logarithmic empirical data provides the derivative at large times, hence an equation with one unknown, k_g. The results of the experiment are shown in Fig. 5. The temperature entering the top of the annulus, $T_w(H, t)$, and the temperature leaving the top of the return pipe, $T_r(H, t)$ are plotted with respect to time. Fig. 6 shows the data in semi-logarithmic form, clearly showing the log-linear behavior at large time.

A value for the volumetric heat capacity ($\rho_g C_{pg}$) is approximated by rearranging Eq. (10) to solve for the thermal diffusivity. Eq. (10) assumes the well to be isothermal in the z-direction, however, the annular flow is non-isothermal. Therefore, the temperature of the well wall at the bottom of the well, $T_g(R_o, 0, t)$ is used in Eq. (15) because it is the only temperature not affected by short-circuiting through the return pipe and by the temperature shift of the heat load from the external heat pump system; hence, it has the closest resemblance to the temperature of the well for the constant heat flux assumption in Eq. (10). However, $T_g(R_o, 0, t)$ data are not measured in the experiment. Only $T_w(H, t)$, $T_r(H, t)$, T_i, Q, and m are empirically obtained. Therefore, $T_g(R_o, 0, t)$ is calculated through approximation. Fig. 7 shows a typical temperature profile down the annulus and up the return pipe. ΔT_{SC} is the temperature increase due to short-circuiting heat transfer from the annulus to the return pipe. Here, ΔT_{HP} is the temperature increase from the external heat pump, shown in Eq. (4). ΔT_G is the temperature increase due to the thermal resistance from the well wall to the point where the temperature in the formation is T_i. T_i is the initial ground formation temperature. Starting at $T_r(H, t)$ in Fig. 7, $T_g(R_o, 0, t)$ is calculated by calculating $T_r(0, t)$ using ΔT_{SC}, and then relating

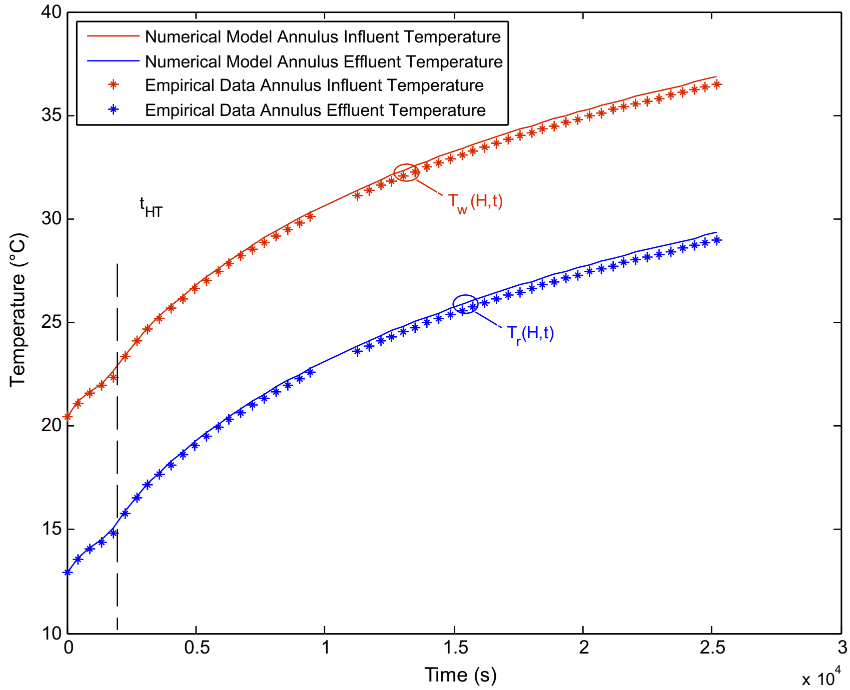

Fig. 5. Validation data for the numerical model using empirical data.

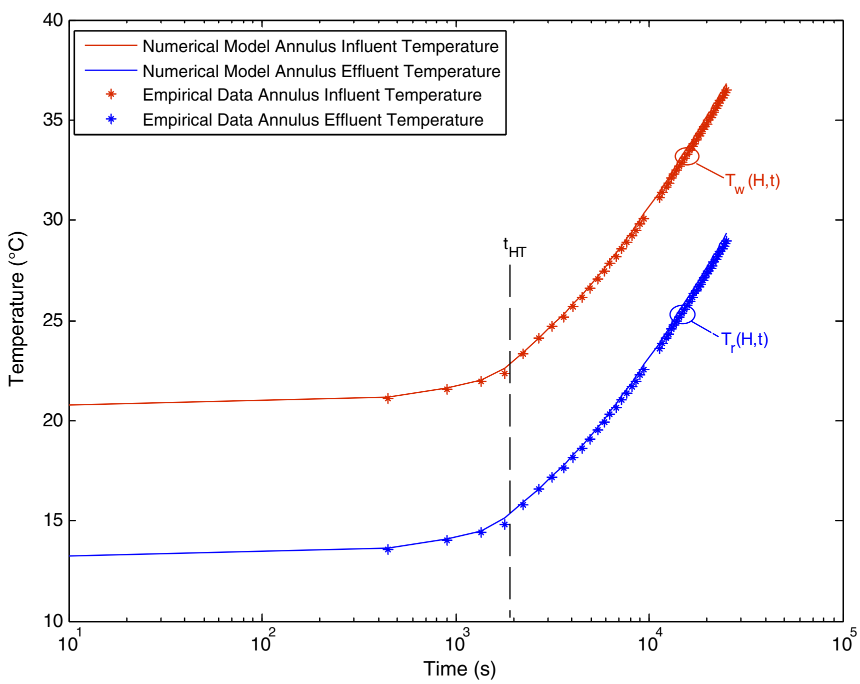

Fig. 6. Validation data in semi-logarithmic form depicting linear profile due to radial conduction.

$T_g(R_o, 0, t)$ to $T_r(0, t)$ through the convective resistance, $1/h_o$. Typically Nusselt numbers are on the order of 100; hence, the convective resistance is negligibly small. Thus with these assumptions, Eq. (10) yields

$$\alpha_g = \frac{k_g}{\rho_g C_{pg}} = \frac{exp(\gamma)R_0^2}{4t} \exp\left(\frac{T_r(H,t) - \Delta T_{sc} - T_i}{Q/4\pi H k_g} - \frac{k_g}{k_w}\frac{1}{Nu}\right) \qquad (15)$$

k_w is the conductivity of water. The experiment showed the ground to have an effective conductivity of 3.2 W/m K (1.856 BTUH/ft°·F) and a thermal diffusivity of $1.1 \cdot 10^{-6}$ m^2/s (1.02 ft^2/day). These properties are within known thermal conductivity and diffusivity ranges for rocks of 2.15–5.38 W/m K, and $0.7 \cdot 10^{-6}$–$3.35 \cdot 10^{-6}$ m^2/s, as tabulated in [16].

4. Numerical model development

In this section a numerical model is developed in order to simulate the transient behavior of the temperature field in the well

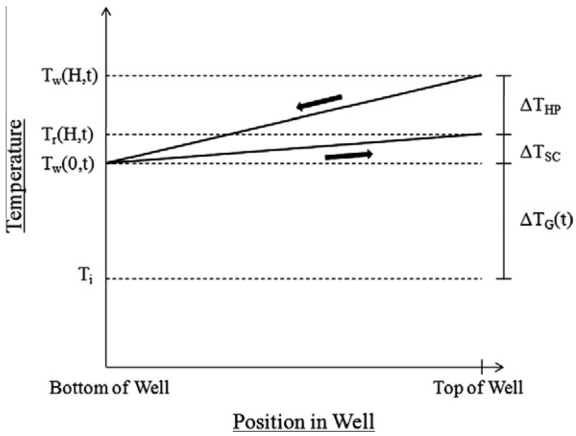

Fig. 7. Temperature profile down the annulus and up the return pipe. The approximate analytical solution parameters are described graphically on the right side.

and surrounding formation. The model will then be used to validate an approximate analytical solution developed in the next section. Despite the fact that the well has a circular cross-section, it was advantageous to base the grid on a Cartesian coordinate system, and Fig. 4 shows how the circular well was converted to an equivalent square well that facilitated its integration into the formation Cartesian grid.

4.1. Internal domain discretization

Fig. 8 shows a coarse grid used to describe the intricacies of the numerical discretization. The discretized energy equation for the ground formation was derived using node 1 from Fig. 8. Fig. 8 shows the subscripts *i*, *j*, and *k* that are used to describe the node position in the *x*, *y*, and *z* directions, respectively. The three governing equations (1), (2), and (12) were discretized into explicit form, as follows:

$$T_w^{t+\Delta t}(k) = (1 - Co_{d,k} - NTU_{1d} - NTU_{2d})T_w^t(k) + NTU_{1d}T_{g,wall}^t(k) + NTU_{2d}T_r^t(k) + Co_{d,k}T_w^t(k-1) \tag{16}$$

$$T_r^{t+\Delta t}(k) = (1 - Co_{u,k} - NTU_{1u})T_r^t(k) + NTU_{1u}T_w^{t+\Delta t}(k) + Co_{u,k}T_r^t(k+1) \tag{17}$$

$$T_g^{t+\Delta t}(i,j,k) = (1 - Fo_1 - Fo_2 - Fo_3 - Fo_4)T_g^t(i,j,k) + \Big(Fo_1T_g^t(i-1,j,k) + Fo_2T_g^t(i+1,j,k) + Fo_3T_g^t(i,j-1,k) + Fo_4T_g^t(i,j+1,k)\Big) \tag{18}$$

$T_{g,wall}$ is the average temperature of the boundary nodes 9–13. The non-dimensional parameters from Eqs. (16)–(18) are a function of position in the domain due to variable node spacing, as follows:

$$Co_{d,k} = \frac{m\Delta t}{\rho_w A_{cd}\Delta z_{k-1}} \tag{19}$$

$$Co_{u,k} = \frac{m\Delta t}{\rho_w A_{cu}\Delta z_k} \tag{20}$$

where *Co* is a so-called Courant number,

$$NTU_{1d} = \frac{4lh_o\Delta t}{\rho_w C_{pw} A_{cd}} \tag{21}$$

$$NTU_{2d} = \frac{2\Delta t}{\rho_w A_{cd} C_{pw} R_{\text{pipe}}} \tag{22}$$

$$NTU_{1u} = \frac{2\Delta t}{\rho_w A_{cu} C_{pw} R_{\text{pipe}}} \tag{23}$$

where *NTU* is a the number of transfer units, a parameter that commonly occurs in heat exchanger analysis, and

$$Fo_1 = \frac{\alpha_g\Delta t}{\Delta x_{i-1}\left(\frac{\Delta x_{i-1}+\Delta x_i}{2}\right)} \tag{24}$$

$$Fo_2 = \frac{\alpha_g\Delta t}{\Delta x_i\left(\frac{\Delta x_{i-1}+\Delta x_i}{2}\right)} \tag{25}$$

$$Fo_3 = \frac{\alpha_g\Delta t}{\Delta y_{j-1}\left(\frac{\Delta y_{j-1}+\Delta y_j}{2}\right)} \tag{26}$$

$$Fo_4 = \frac{\alpha_g\Delta t}{\Delta y_j\left(\frac{\Delta y_{j-1}+\Delta y_j}{2}\right)} \tag{27}$$

where *Fo* is the Fourier number. Here Δx_i, Δx_{i-1}, Δy_j and Δy_{j-1} are the inter-node spacing shown in Fig. 4.

The Cartesian domain introduces a problem in matching the Cartesian grid to the circular well boundary. This is addressed by creating an apparent square well as shown in Fig. 4. The cross-sectional area of the well is chosen such as to conserve $Co_{d,k}$, the scaled residence time for flow to pass through the annulus region encompassed by node *k*. NTU_{1d}, a function of the convection thermal resistance and the only other non-dimensional parameter affected by the square well assumption, is adjusted by changing the perimeter to $4l$ instead of $2\pi R_o$ in Eq. (21). The convection thermal resistance from the well to the ground is negligible in the cylindrical well, therefore the square well assumption has a minimal effect on the final solution.

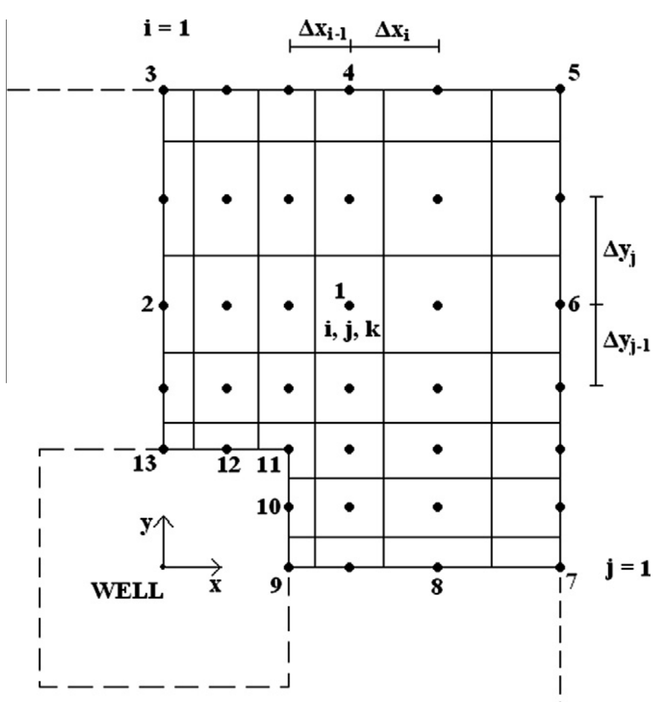

Fig. 8. Coarse grid of the numerical domain for the ground formation connected to the annular flow domain.

4.2. Boundary discretization

There are two boundary conditions for the annular flow and return pipe flow. Eq. (28), an energy balance on the external heat load, produces the first condition. Eq. (29) states that at the well bottom, the temperature of the down-coming annulus flow is the same as the temperature of the return flow at that point

$$T_w^{t+\Delta t}(1) = T_r^t(1) + \frac{Q}{mC_{pw}} \tag{28}$$

$$T_r^{t+\Delta t}(N) = T_w^{t+\Delta t}(N) \tag{29}$$

Here N is the number of nodes in the z-direction.

There are 6 boundary conditions for the ground formation, four on the sides of the ground domain (nodes 2, 4, 6, and 8 in Fig. 8), and two on the annulus-ground interface (nodes 10 and 12 in Fig. 8). The numerical model also required the discretization of six corner nodes (nodes 3, 5, 7, 9, 11, and 13 in Fig. 8). As an example, Fig. 9 shows the discretization of node 10 producing:

$$\begin{aligned}(\rho_g C_{pg})\frac{\Delta x_{i-1}}{2}\left(\frac{\Delta y_{j-1}+\Delta y_j}{2}\right)\Delta z_k \frac{\Delta T_g(i,j,k)}{\Delta t} &= h_o\left(\frac{\Delta y_{j-1}+\Delta y_j}{2}\right)\Delta z_k (T_w^{t+\Delta t}(k) - T_g^t(i,j,k)) + k_g\frac{\Delta x_{i-1}}{2}\Delta z_k \\ &\times \frac{(T_g^t(i,j+1,k) - T_g^t(i,j,k))}{\Delta y_j} + k_g \frac{\Delta x_{i-1}}{2}\Delta z_k \\ &\times \frac{(T_g^t(i,j-1,k) - T_g^t(i,j,k))}{\Delta y_{i-1}} + k_g\left(\frac{\Delta y_{j-1}+\Delta y_j}{2}\right)\Delta z_k \\ &\times \frac{(T_g^t(i+1,j,k) - T_g^t(i,j,k))}{\Delta x_{i-1}}\end{aligned} \tag{30}$$

The first term in Eq. (30) is the energy storage term for the node volume, the second term is the convective term from the annulus, and the remaining terms characterize the conduction to all adjacent nodes in the x and y directions. Taking the limit as Δx approaches

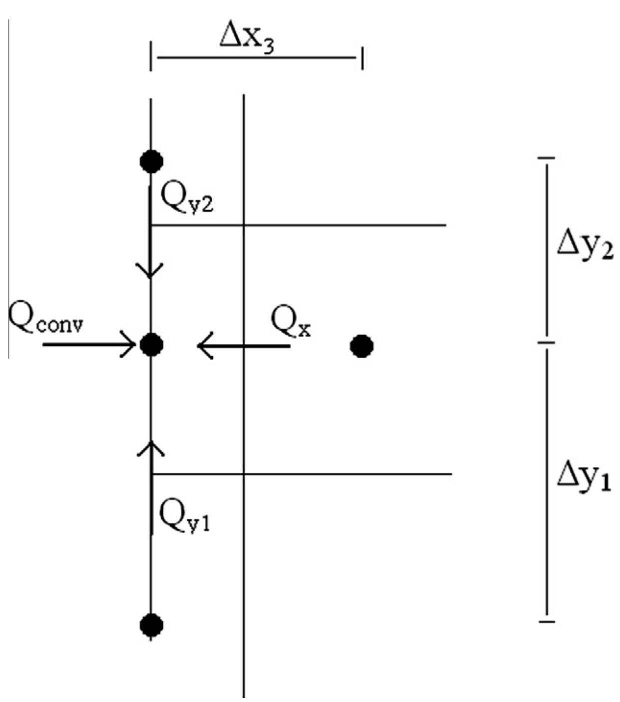

Fig. 9. Derivation of the equation for node 10 on the surface of the square well in Fig. 8.

dx reduces Eq. (30) to the boundary condition in Eq. (8), modified for Cartesian coordinates. Solving for the temperature at the next time step produces:

$$\begin{aligned}T_g^{t+\Delta t}(i,j,k) = T_g^t(i,j,k) &+ \frac{2h_o\Delta t}{(\rho_g C_{pg})\Delta x_{i-1}}(T_w^{t+\Delta t}(k) - T_g^t(i,j,k)) \\ &+ \frac{2\alpha_g \Delta t}{(\Delta y_{j-1}+\Delta y_j)\Delta y_j}(T_g^t(i,j+1,k) - T_g^t(i,j,k)) \\ &+ \frac{2\alpha_g \Delta t}{(\Delta y_{j-1}+\Delta y_j)(\Delta y_{j-1})}(T_g^t(i,j-1,k) - T_g^t(i,j,k)) \\ &+ \frac{2\alpha_g \Delta t}{(\Delta x_{i-1})^2}(T_g^t(i+1,j,k) - T_g^t(i,j,k))\end{aligned} \tag{31}$$

The remaining five boundary nodes as well as the six corner nodes are solved in a similar fashion. Table 1 shows the parameters used in three simulations using the numerical model. The parameter values were modified in each simulation to ensure the robustness of the numerical code. These simulations are used to validate the analytical model in the following section.

5. Approximate analytical model development

In this section, an approximate analytical solution is derived. Although the resulting analytical model is not exact, it allows rapid computations that are reasonably accurate. The more accurate numerical model is used to validate the approximate analytical model. The most important temperatures in this analysis are the temperature of water exiting the return pipe and the temperature of the water entering the annulus, as they influence the coefficient of performance of the heat pump system to which the well is connected. As such, the analytical solution is derived by focusing on the water entering the annulus, studying its interaction with the ground formation and return pipe, and solving for the exiting temperature. The solution is scaled to the initial ground temperature, T_i, where at time $t = 0$, the effluent temperature $T_r(H, t)$ is T_i since the standing water column is initially in equilibrium with the formation.

5.1. Early times, $t < t_{HT}$

For short time periods, $t < t_{HT}$, the thermal response is governed by the thermal mass of the water in the well. For typical wells, this time regime is not important in the analysis, but the temperature at time t_{HT} is. According to [11], the profile of the temperatures in the annulus and return pipe may be approximated as linear. Referring to Fig. 7, at time t_{HT}, $T_w(0, t)$ and $T_r(0, t)$ are approximately T_i, and the temperatures at the inlet of the annulus $T_w(H, t)$ and the outlet of the return pipe $T_r(H, t)$ are obtained by calculating the temperature change using the approximate linear profiles. The temperature change through the return pipe is approximated by

Table 1
Numerical simulation parameters.

Parameter	Numerical simulation #1	Numerical simulation #2	Numerical simulation #3
Heat load, W (tons)	26,375 (7.5)	10,000 (2.93)	75,000 (21.9)
Ground conductivity, W/m K	3.2	3.2	5
Flow rate, kg/s (GPM)	1.58 (25)	2.0186 (32)	4.73 (75)
Length between wells, m	Infinite	0.7	2.07
Well diameter, m (in)	0.1524 (6)	0.1016 (4)	0.3048 (12)
Convection coeff., W/m^2 K	500	2000	400
Pipe resistance, mK/W	0.114	0.114	0.114
Depth of well, m	220	75	125

an energy balance on the short circuiting of heat through the return pipe, Eq. (32)

$$\Delta T_{SC} = T_r(H,t) - T_r(0,t) = \frac{1}{2}\Delta T_{HP}\frac{2H}{R_{pipe} m C_{pw}} = \frac{1}{2}\Delta T_{HP} NTU \tag{32}$$

NTU and R_{pipe} are given in Eq. (3). The temperature change up the annulus is approximated by an energy balance across the external heat source. The water exiting the return pipe enters the annulus with a different temperature, also shown in Eq. (4), calculated by

$$\Delta T_{HP} = T_w(H,t) - T_r(H,t) = \frac{Q}{mC_{pw}} \tag{33}$$

The integration of Eqs. (32) and (33) are illustrated in Fig. 7. The influent temperature to the annulus and the effluent temperature of the return pipe at t_{HT}, scaled to the initial temperature of the formation are

$$\begin{aligned} T_w(H,t_{HT}) - T_i &= (T_w(H,t_{HT}) - T_r(H,t_{HT})) + (T_r(H,t_{HT}) - T_r(0,t_{HT})) \\ &= \Delta T_{HP} + \Delta T_{SC} \end{aligned} \tag{34}$$

$$T_r(H,t_{HT}) - T_i = T_r(H,t_{HT}) - T_r(0,t_{HT}) = \Delta T_{SC} \tag{35}$$

Scaling t_{HT} produces a Fourier number:

$$Fo_{R,HT} = \frac{\alpha_g t_{HT}}{R_o^2} = \frac{\alpha_g}{R_o^2}\left(\frac{\rho_w V_{return} + \rho_w V_{annulus}}{m}\right) \tag{36}$$

The linear profile assumption fails as $Fo_{R,HT}$ increases because the characteristic residence time for water through the well approaches the characteristic time for conduction through the formation; hence, the thermal profile deviates from the constant heat flux assumption. As such, this aspect of the analytical solution is site specific, but for typical standing column well geometries, thermal properties, and operating conditions, the linear assumption is generally valid.

5.2. Intermediate times, $t_{HT} < t < t_R$

After the well water is thermally saturated, the thermal response transitions into a regime dominated by heat exchange into the formation. Mathematically it is represented by a semi-infinite radial problem with a constant heat flux on the well wall. The well temperatures are influenced by heat transfer to the ground formation in addition to ΔT_{HP} and ΔT_{SC}. The additional ground formation thermal resistance from the well to the undisturbed formation temperature incorporates the change in temperature of the entire well associated with the heat transfer through the formation. This is shown in Fig. 7 by $\Delta T_G(t)$. It is derived from Eq. (10) and includes the convective heat transfer coefficient from the annulus flow to the ground, h_o

$$\Delta T_G(t) = q_r\left[\frac{R_o}{2k_g}\ln\left(\frac{4Fo_R}{\exp(\gamma)}\right) + \frac{1}{h_o}\right] \tag{37}$$

where

$$Fo_R = \frac{\alpha_R t}{R_o^2} \tag{38}$$

and

$$q_r = \frac{Q}{2\pi R_o H} \tag{39}$$

This equation is valid for $Fo_R > 5$ [15]. The final approximate $T_w(H,t)$ and $T_r(H,t)$ are therefore:

$$T_w(H,t) = T_i + \Delta T_{HP} + \Delta T_{SC} + \Delta T_G(t) \tag{40}$$

$$T_r(H,t) = T_i + \Delta T_{SC} + \Delta T_G(t) \tag{41}$$

The water temperature in Eq. (40) is scaled as follows:

$$\theta = \frac{T - T_i}{Q/2\pi H k_g} \tag{42}$$

producing

$$\theta_w = NTU_g\left(1 + \frac{1}{2}NTU\right) + \frac{1}{2}\ln\left(\frac{4Fo_R(t)}{e^\gamma}\right) + \sigma\frac{1}{Nu} \tag{43}$$

Taking the derivative of Eq. (43) with respect to the natural logarithm of time, as suggested by the analysis used in Eq. (14) for empirical calculation of the ground formation effective properties gives

$$\frac{d\theta_w}{d\ln(Fo_R)} = \frac{1}{2} \tag{44}$$

This analysis yields the same result through scaling Eq. (41) since NTU_g and NTU are constants. For any numerical simulation, the results should scale in a similar fashion. Eq. (43) is validated in Section 6 by comparing the numerical simulations to the behavior predicted by Eq. (44) for the approximate analytical solution.

5.3. Late times, $t > t_r$

The radially dominated behavior remains until the thermal front propagates to the axis of symmetry between adjacent wells in the infinite line. Shown in Fig. 3, for $t > t_R$, the thermal behavior near the well is radial heat flow while the farfield exhibits planar heat flow. Scaling to the initial temperature T_i in the farfield, the problem is assumed to transition to a planar solution from the radial solution at a time t_R. This transition time, t_R, is approximated in Eq. (11). In addition to the radial thermal resistance described in Eq. (37), another apparent thermal resistance affects the heat flow to the formation. This thermal resistance may be viewed as occurring across a temperature drop $\Delta T_p(t)$ as shown in Fig. 10. $\Delta T_p(t)$ is derived from a constant heat flux semi-infinite planar solution evaluated at $y = 0$ from Incropera and Dewitt [16] and is given by

$$T_g(x,0,t) - T_i = 2\frac{q_y}{k_g}\left(\frac{\alpha_g t}{\pi}\right)^{1/2} \tag{45}$$

where

$$q_y = \frac{Q}{2L_f H} \tag{46}$$

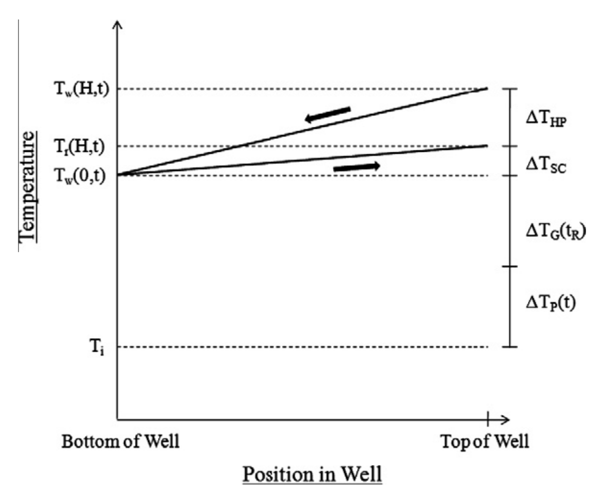

Fig. 10. Temperature profile down the annulus integrating the additional thermal resistance due to planar flow.

Eq. (45) is a solution for a planar semi-infinite half space and hence assumes isothermality with respect to the x-direction. Fig. 3 illustrates non-isothermal behavior in the x-direction for $t > t_R$; however, this non-isothermal behavior is assumed to be captured by $\Delta T_G(t_R)$ and has little effect on $\Delta T_p(t)$. In order to apply Eq. (45) from the transition time t_R forward to calculate $\Delta T_p(t)$, a temporal integration yields:

$$\Delta T_P(t) = \int_{t_R}^{t} \frac{dT_g}{ds} ds = \int_{t_R}^{t} \frac{1}{\sqrt{\pi}} \frac{Q}{L_f H k_g} \left(\frac{\alpha_g}{s}\right)^{1/2} ds$$
$$= 2\sqrt{\pi} \frac{Q}{2\pi H k_g} \left(Fo_L^{1/2} - \frac{R_o}{L_f} Fo_R(t_R)^{1/2} \right) \tag{47}$$

where

$$Fo_L = \frac{\alpha_g t}{L_f^2} \tag{48}$$

and Fo_R is given in Eq. (36). L_f is the length over which the constant heat flux is applied. Fig. 11 was produced using the numerical model operating under the parameters of the field experiment. It shows the temperature response parametric on L_f/R_o and scaled to the square root of time as deduced from Eqs. (45) and (47). Fig. 11 shows that at large well spacing ($L_f/R_o = 16, 32, 64$) the well temperature does not scale as $t^{1/2}$, indeed it scales as $\ln(t)$ for radial heat flow in the formation, as seen in Fig. 6. As L_f/R_o decreases, the response transitions to $t^{1/2}$ behavior for one-dimensional planar heat flow, as clearly seen for $L_f/R_o = 2.2$, and 4. Scaling Eq. (47) as per Eq. (42) and taking the derivative with respect to the square root of Fourier number defined in Eq. (48) yields a constant:

$$\frac{d\theta}{d(Fo_L)^{1/2}} = 2\sqrt{\pi} \approx 3.55 \tag{49}$$

In the late time regime the behavior suggested by Eq. (49) provides a measure by which to validate the numerical solution. One final detail involves the spatial distance on the surface of the half space upon which to apply the constant heat flux condition. The simplest assumption is that $L_f = L$, meaning the distance between wells (shown in Fig. 1) is where the constant heat flux boundary condition is applied; however, the well is square and protrudes into the domain. Therefore, an extra length l (shown in Fig. 4) may be added to produce $L_f = L + \mathrm{l}$. Since the length L_f is unknown, these two definitions bound the solution. When $L \gg \mathrm{l}$, L_f approaches L.

The analytical temperature for times greater than t_R is therefore defined as

$$T_w(H,t) = T_i + \Delta T_{\mathrm{HP}} + \Delta T_{\mathrm{SC}} + \Delta T_G(t_R) + \Delta T_P(t) \tag{50}$$

and in non-dimensional form:

$$\theta_w = NTU_g \left(1 + \frac{1}{2} NTU\right) + \frac{1}{2} \ln\left(\frac{4Fo_R(t_R)}{e^{\gamma}}\right) + \sigma \frac{1}{Nu} + 2$$
$$\times \sqrt{\pi} \left(Fo_L^{1/2} - \frac{R_o}{L_f} Fo_R(t_R)^{1/2} \right) \tag{51}$$

Here,

$$NTU_g = \frac{2\pi H k_g}{m C_{pw}} \tag{52}$$

and

$$\sigma = \frac{k_g}{k_w} \tag{53}$$

5.4. Model validation

Using the empirical data from [14] the numerical model was validated as shown in Figs. 5 and 6. The slope of Fig. 6 is constant as time increases and allows for the calculation of the thermal conductivity as described in Eq. (14). The numerical model is clearly accurate in capturing the well thermal response. Both the well data and the numerical model clearly show the early well dominated behavior for $t < t_R$ followed by the formation dominated behavior.

The empirical results are for an isolated well while the numerical solution allows for an infinite line of wells with varying spacing. Therefore, in Figs. 12 and 13, the numerical solution is compared with the analytical parameters described by Eqs. (44) and (49). Table 1 shows the parameters used for three model

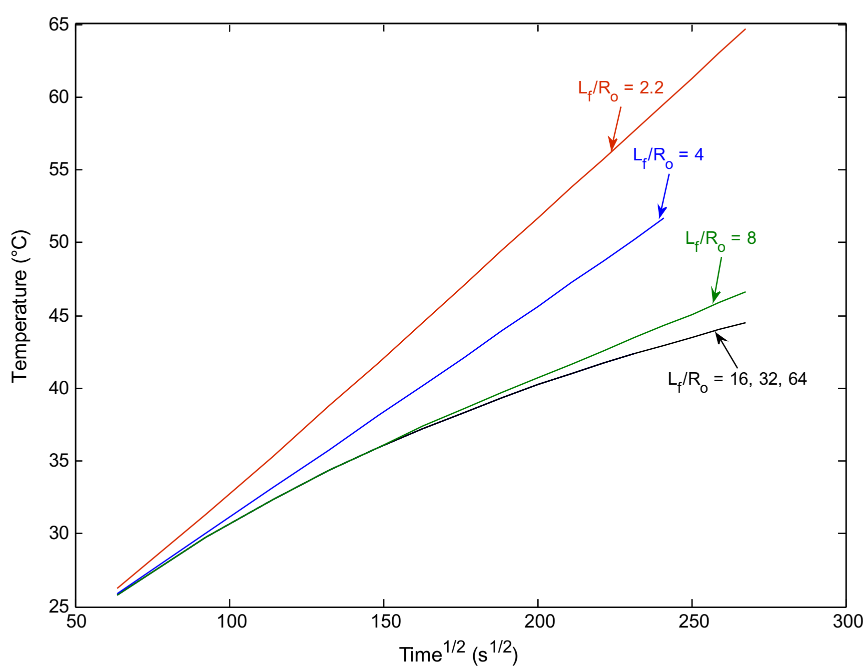

Fig. 11. Influent well temperature calculated from the numerical model using the parameters from the field experiment, varying L_f/R_o, and scaling to the square root of time.

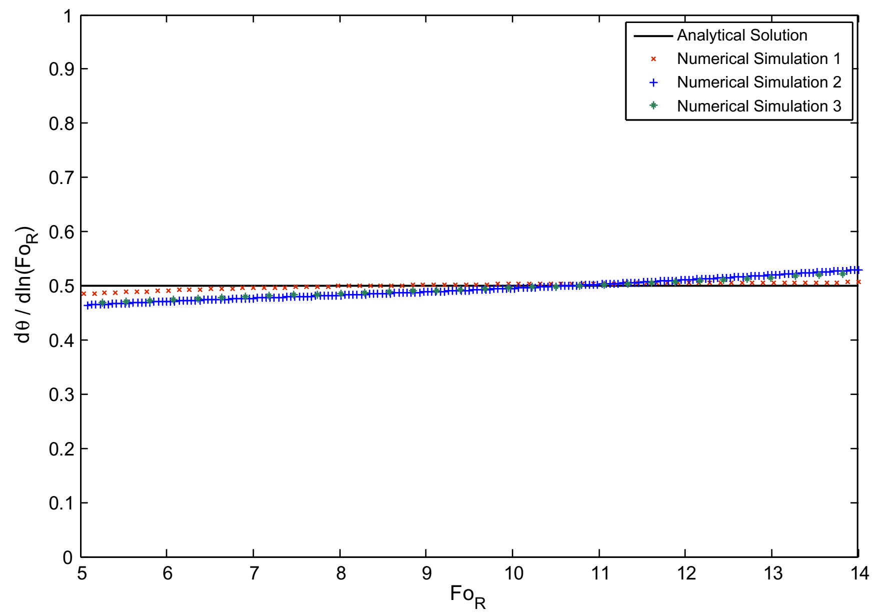

Fig. 12. Comparison of numerical simulations described in Table 1 to the analytical constant heat flux semi-infinite solid temperature gradient obtained in Eq. (44).

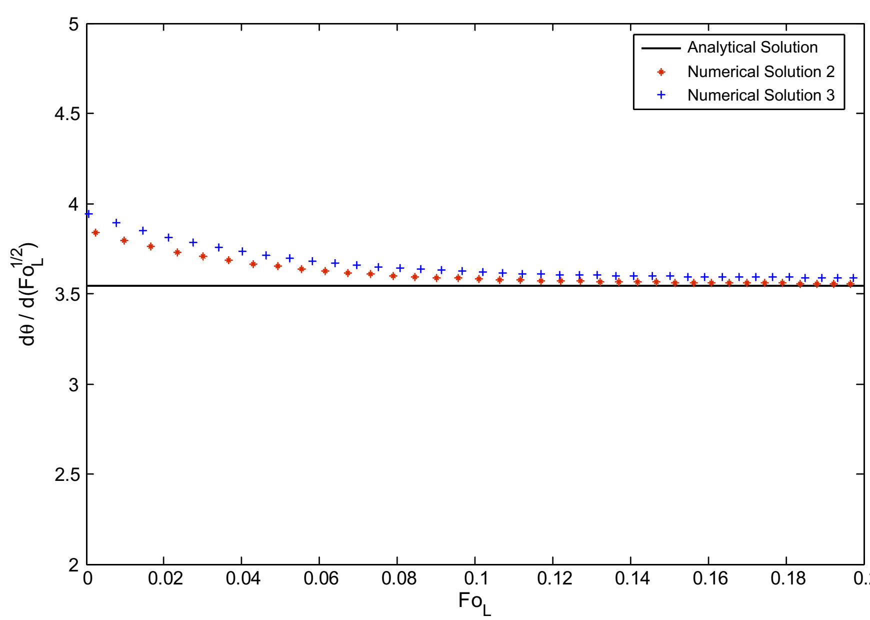

Fig. 13. Comparison of the numerical simulations to the analytical temperature gradient for a semi-infinite planar solution obtained in Eq. (49).

simulations. Fig. 12 shows that $d\theta/d\ln(Fo_R)$ is not a constant, but weakly time dependent; the approximate analytical model however introduces little error by the assumption of linear well temperature profiles. As time increases, Fig. 13 shows that the numerical solutions approach the thermal response predicted by the analytical solution for planar heat flow. Figs. 14 and 15 show the correlation of the approximate analytical solution, with and without the planar component, to the results of numerical simulations 2 and 3, respectively. The errors in Figs. 14 and 15 are approximately 2–3 °C. This error is caused by, first, the assumption of a linear profile for the annulus and return pipe, and second, the assumption that there is a fixed transition time rather than a transition time range. As shown in Fig. 13, the numerical simulations approach the value from Eq. (50) after the transition time calculated from Eq. (11). These errors are generally negligible for the analysis of the coefficient of performance.

5.5. Thermal degradation factor

Eq. (51) when divided by Eq. (43) for the radial analytical solution produces a degradation factor that compares the temperature response of the infinite line of wells to that of an isolated well,

$$\eta\left(\frac{R_o}{L_f},Fo_L,NTU_g,NTU,\frac{\sigma}{Nu}\right)=\frac{(T_w(H,t)-T_i)_{lineofwells}}{(T_w(H,t)-T_i)_{isolatedwell}}=\frac{\theta_{w,p}}{\theta_{w,r}}$$
$$=\frac{[NTU_g(1+\frac{1}{2}NTU)+\frac{1}{2}\ln(\frac{4Fo_R(t_R)}{e^{\gamma}})+\sigma\frac{1}{Nu}+2\sqrt{\pi}(Fo_L^{1/2}-\frac{R_o}{L_f}Fo_R(t_R)^{1/2})]}{[NTU_g(1+\frac{1}{2}NTU)+\frac{1}{2}\ln(\frac{4Fo_R(t)}{e^{\gamma}})+\sigma\frac{1}{Nu}]} \quad (54)$$

Here $\theta_{w,p}$ is the scaled temperature response for a line of wells, and $\theta_{w,r}$ is the scaled temperature response for an isolated well. Using Eqs. (11) and (38), the non-dimensional time $Fo_R(t_R)$ is related to R_o/L by

$$Fo_R(t_R) = \frac{e^{\gamma}}{4}\left[\frac{1}{4}\left(\frac{L_f}{R_o}\right)^2 - \frac{L_f}{R_o} + 1\right] \tag{55}$$

Also, Eqs. (38) and (48) are related by

$$Fo_R(t) = \left(\frac{L_f}{R_o}\right)^2 Fo_L(t) \tag{56}$$

Applying Eqs. (55) and (56) and taking the limit as the mass flow rate, m, approaches infinity simplifies Eq. (54) to:

$$\eta\left(\frac{R_o}{L_f}, Fo_L\right) = \frac{\ln\left(\frac{1}{4}\left(\frac{L_f}{R_o}\right)^2 - \frac{L_f}{R_o} + 1\right) + 4\sqrt{\pi}\left(Fo_L^{1/2} - \left[\frac{e^{\gamma}}{4}\left(\frac{R_o^2}{L_f^2} - \frac{R_o}{L_f} + \frac{1}{4}\right)\right]^{1/2}\right)}{\ln\left(\frac{4L_f^2}{e^{\gamma}R_o^2}Fo_L\right)} \tag{57}$$

Because the thermal response in the intermediate and late time regimes is dominated by the formation, the implication of assuming

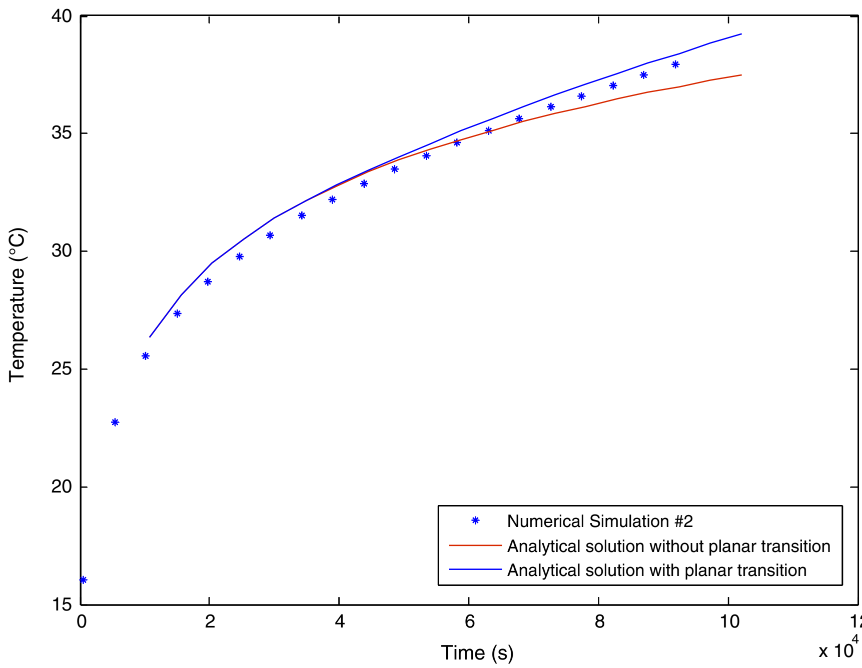

Fig. 14. Numerical simulation #2 results compared to the approximate analytical solution showing the deviation from radial behavior to planar behavior.

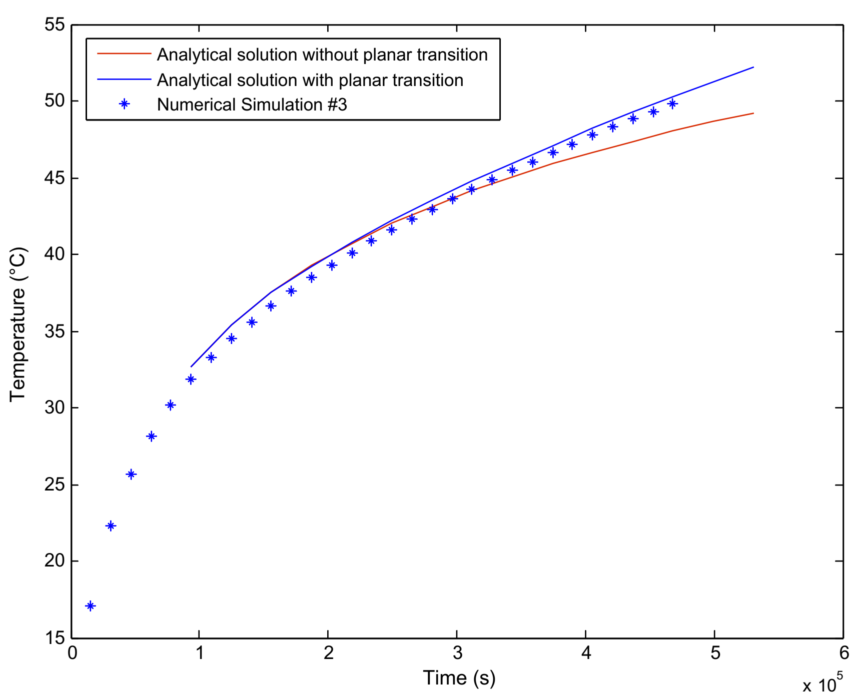

Fig. 15. Numerical simulation #3 results compared to the approximate analytical solution showing the deviation from radial behavior to planar behavior.

an infinite well flow rate is to ignore the wellbore contribution to the thermal response; as previously shown, this is a valid assumption. Following Witte et al. [15], Eq. (57) has limitations similar to those described in Eq. (37). From [15], Fo_R in Eqs. (55) and (56) must be larger than 5 for the solution to be valid. These limitations on Eqs. (55) and (56) therefore are:

$$5 < \frac{e^{\gamma}}{4}\left[\frac{1}{4}\left(\frac{L_f}{R_o}\right)^2 - \frac{L_f}{R_o} + 1\right] \tag{58}$$

and

$$Fo_L(t) > 5\left(\frac{R_o}{L_f}\right)^2 \tag{59}$$

Solving for L_f/R_o in Eq. (58), the minimum L_f/R_o that can be analyzed in this section is approximately 8.7, and, from Eq. (59), the minimum Fo_L, using $L_f/R_o = 8.7$, is approximately 0.066. Since typical wells are spaced on the order of meters and well radii are ~0.1 m, typical L_f/R_o values are approximately 10–100. Therefore, the range of validity of the approximate model does not impede its utility for practical calculations.

6. Results and discussion

In this section, the impact of well spacing on the coefficient of performance of an idealized two temperature heat pump is presented, as well as a description of the variation of the coefficient of performance with the non-dimensional parameters in Eqs. (54) and (57).

6.1. Degradation factor

The degradation factor, η, from Eq. (57) is shown in Fig. 16. It is interesting to note that the results in Fig. 11 for $L_f/R_o = 2$, 4, and 8, when divided by the results for $L_f/R_o = 16$, 32, and 64 produce a plot similar to Fig. 16; $L_f/R_o = 16$, 32, and 64 have not transitioned to planar heat flow, and only demonstrate radial heat flow characteristics. Fig. 16 is constructed under the assumptions presented for Eq. (57) and varies with changes in σ/Nu, NTU, and NTU_g shown in Eqs. (51) and (54). Including any of these parameters decreases the results in Fig. 16 towards $\eta = 1$; hence, a scaled Fig. 11 is similar but not equivalent to Fig. 16. Since Fig. 16 is the comparison of the thermal response of a line of wells to that of an isolated well, driving the response towards $\eta = 1$ is advantageous. When σ/Nu, NTU, and NTU_g are included, the scaled temperature from Eq. (44) for an isolated well increases and offsets any decrease to η. Therefore, it is advantageous to minimize σ/Nu, NTU, and NTU_g.

Not included in this analysis is the hydraulic analysis of the well which for the assumption of $m \to \infty$ drives the pressure drop, and hence pump work input upwards, towards infinity. This is the classic heat transfer problem, compromising pressure drop for heat transfer.

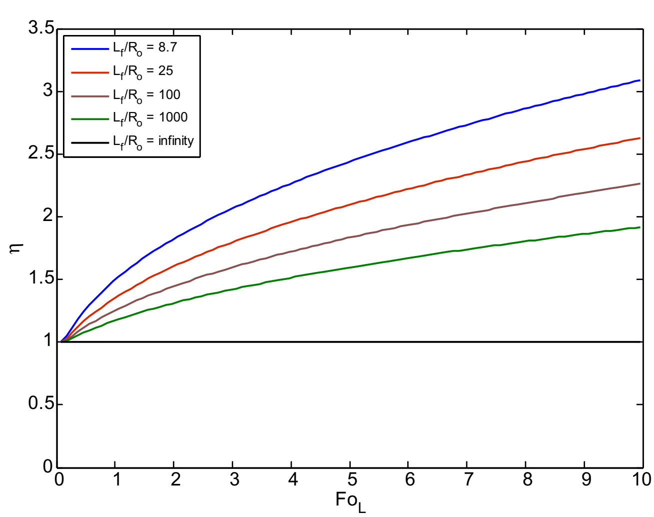

Fig. 16. Degradation factor from Eq. (57) as a function of Fo_L and well spacing L_f/R_o. The infinite flow rate assumption leads to σ/Nu, NTU, and NTU_g equaling zero.

6.2. Analysis of the coefficient of performance

The reversible COP for an ideal two temperature heat pump in refrigeration and heating mode, respectively, are:

$$\mathrm{COP}_R = \frac{T_C}{T_H - T_C} = \frac{\left[\theta_B + \frac{T_i}{Q/2\pi H k_g}\right]}{\eta\theta_{w,r} - \theta_B}, \quad \theta > 0 \tag{60}$$

$$\mathrm{COP}_H = \frac{T_H}{T_H - T_C} = \frac{\left[\theta_B + \frac{T_i}{Q/2\pi H k_g}\right]}{\theta_B - \eta\theta_{w,r}}, \quad \theta < 0 \tag{61}$$

Here θ_B is the non-dimensional building temperature and is typically close to zero. $\theta_{w,r}$ is the temperature response for an isolated well as stated in Eq. (55). In refrigeration mode, because η is larger than unity and the heat load, Q, is positive, the denominator increases; hence, the coefficient of performance decreases. Since θ_B is close to zero, the numerator and denominator in Eq. (62) are typically negative. In heating mode, because η is larger than unity and the heat load, Q, is negative, the denominator decreases; hence, the coefficient of performance decreases for both modes with $\eta > 1$. Assuming that $\theta_B = 0$, meaning the building is kept at the same temperature as the initial ground temperature, the coefficient of performance for both heating modes become equivalent and, when Eqs. (61) and (62) are divided by their ideal values at $\eta = 1$, yields

$$\frac{\mathrm{COP}(\eta)}{\mathrm{COP}(\eta = 1)} = \frac{1}{\eta} \tag{62}$$

The COP ratio is plotted in Fig. 17 versus time and well spacing. There is a noticeable reduction in the COP as time increases for finite well spacing. These results have practical significance because the utilization of ground coupled heat pumps depends on the high COP obtained by the system. If well fields are improperly designed, the COP gain from using ground sourcing may be offset by the COP

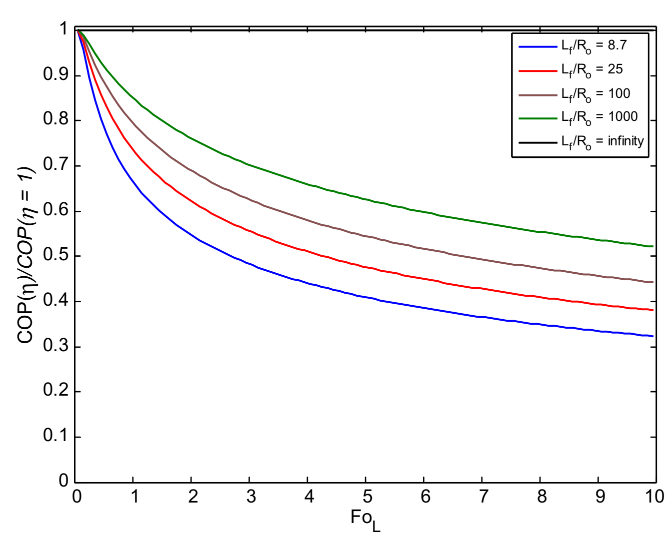

Fig. 17. Coefficient of performance ratio from Eq. (63) as a function of Fo_L and well spacing L_f/R_o. The infinite flow rate assumption leads to σ/Nu, NTU, and NTU_g equaling zero.

degradation due to thermal encroachment from adjacent wells. For example, a line of wells of 0.1 m radius, spaced 5 m apart, with a thermal diffusivity of $3.35 \cdot 10^{-6}$ m^2/s, running for 1000 h (a typical season), would exhibit a COP reduction of 12% at the end of the season compared to an isolated well.

7. Conclusions

Design of standing column well fields currently has no official standards and relies on the experience of the designer for quality. The results obtained in this paper condense the design problem to a simple analysis of the operating parameters of the well field. A thorough analysis of an infinite line of wells was produced using a numerical model, empirical data, and an approximate analytical solution.

The numerical model had three transient multi-dimensional explicit equations connected by transient boundary conditions. The first and second equations modeled the convection heat transfer from flow in the annulus and return pipe, respectively, and the third modeled conduction heat transfer through the ground formation. The annulus flow was physically in contact with the return pipe flow at the boundaries in the axial direction, and thermally in contact with the return pipe flow and the ground formation in the radial direction.

An experiment was performed on an actual well with a simulated building heat load. This empirical data validated the numerical model for early times. Analytical theory validated the numerical model for later times. The heat transfer coefficients used in the numerical model were derived from common heat transfer correlations for turbulent flow. The ground formation thermal conductivity was calculated using a standard log-linear regression estimation method with the empirical data, and the thermal diffusivity was obtained using a manipulation of a constant heat flux semi-infinite radial solution with the empirical data.

An approximate analytical model was developed by separating the problem into three time regimes. In the first regime, the thermal response of the well was characterized by the thermal capacitance of the well water until t_{HT} when thermal saturation was assumed to occur. t_{HT} was calculated analytically using the residence time of water flowing through the well. The second regime, $t_{\mathrm{HT}} < t < t_R$, was governed by the thermal response of a semi-infinite radial domain by a constant heat flux on an inner radius equal to the radius of the well. This regime remained until the transition time, t_R, defined as the time required for the thermal front to encroach on adjacent wells. The third regime, $t > t_R$, was governed by the thermal response of a constant heat flux boundary condition applied on a 1-D semi-infinite planar domain. The approximate analytical solution was validated against the numerical solution and used to examine the reduction in the coefficient of performance of an infinite line of wells operating concurrently with respect to time and well spacing. The results are:

1. The design of fields of wells to minimize thermal encroachment is important for providing maximum efficiency for heating and cooling systems.
2. The reversible coefficient of performance for transferring heat from a facility to the ground is a function of the non-dimensional variables L_f/R_o, Fo_L, σ/Nu, NTU, and NTU_g.
3. Assuming an infinite flow rate causes σ/Nu, NTU, and NTU_g to approach zero simplifying the problem to a function of L_f/R_o and Fo_L. This assumption allows the problem to be characterized by a single plot (Fig. 17), which quantifies the decrease in the reversible coefficient of performance as well spacing decreases and time increases.
4. Fig. 17 is proposed as a standard for proper design of well fields with minimum degradation from thermal encroachment.

For future consideration, the study of the interaction of two wells operating concurrently, and the study of a field of wells operating concurrently should be considered and compared to Fig. 17. The two well model would be the upper limit to the problem of a finite line of wells where the central wells will approach the infinite line model, while the wells on the extremities will approach the two well model. These solutions will be used to produce a field manual for well field design.

Acknowledgments

We wish to acknowledge ARB Geowell Systems for supplying the empirical data for this analysis. In addition, we acknowledge the valuable assistance of Dr. Gerard Jones on the numerical model development.

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